V.F.S. v2.0 (Open-Gate) — Core Compendium

Expanded Core Formulas, Parameters, and Sophianic Folding Feedback

with the 1.0 / appendix restoration patch merged in

Scope

This file collects the expanded core Open-Gate formulas of V.F.S. v2.0 and preserves the original index-derived structure. The Sophianic Folding layer is integrated as a later Open-Gate refinement rather than as a replacement for the existing core. Formulas that belong specifically to `appendix.html` or `lyapunov.html` remain excluded unless needed to explain the core feedback structure. Graphs, numerical plots, and visual phase diagrams are also excluded. The governing distinction is:

$$\text{V.F.S. v2.0 Open-Gate:}\ \dot{\lambda} = -\dot{\sigma} + I_{\text{gate}}, \qquad \sigma + \lambda = C_0 + G_{\text{recepta}}(t).$$

The closed Microcosm law $\dot{\lambda} = -\dot{\sigma},\ \sigma + \lambda =$ const is only the limit $\zeta_0 = 0$.

▶ [merged] Paragraphs and sections marked ▶ are *added* by the restoration patch (from 1.0 / Appendices I, IV), with all symbols written in the 2.0 convention. *Nothing in the original core is removed.* Two items 2.0 deliberately superseded are *not* added and are flagged where relevant: the Fading-Impulse $\mathcal{P}(t)$/First-Call term, and the closed u_{eff} amplifier route. Blocks marked ▷ Math. add a mathematical reading of what each construction *does* (no new modelling claims).

1 Core State Variables

Symbol	**Name**	**Meaning in V.F.S.**
$V(t)$	Voluntas	Will; inner orientation and willing.
$F(t)$	Factum	Action; embodied execution of will.
$\sigma(t)$	Singular Resistance	Resistance, deformation, or singular obstruction.
$\lambda(t)$	Sophia	Wisdom; transformed resistance plus received grace in Open-Gate.
$P(t)$	Pleroma	Dynamic fullness of the Vessel.
$S(t)$	Being / Sum / Gignesthai	First derivative of Pleroma; rate of becoming.
$\Omega_P(t)$	Brim	Dynamic capacity / rim of the Vessel.
$H_K(t)$	Gate memory	Acquired receptivity; the learned openness of the gate.
$W(t)$	Wound memory	Vulnerability / wound-memory layer paired with H_K.
$q_{\text{fold}}(t)$	Folding amplitude	Morphological shape-mode of the Vessel; zero in the old form, nonzero in Sophianic Folding.
$Q_{\text{fold}}(t)$	Folding intensity	Nonnegative intensity of folding, usually $Q_{\text{fold}} = q_{\text{fold}}^2$.
$\lambda_{\text{form}}(t)$	Embodied Sophia	Portion of Sophia stabilized into Vessel-form rather than remaining as available flow.

▷ Plain. In plain words. V is how strongly you intend (Will), F how much you actually do (Action), σ your inner resistance or hardship, λ the wisdom (Sophia) you have gathered, and Ω_P your capacity to hold being. Everything that follows is built from these.

2 Core Parameters

Param.	Name	Role / condition
ε	Manere seed	Keeps u positive even when V, F vanish; sub-critical Manere condition $0 < \varepsilon < \Lambda_c^2$ keeps the passive branch in Collapse rather than Transformation.
γ	Resistance pressure	Drives growth of σ when synergy is below threshold.
δ	Synergic cutting strength	Converts synergy into resistance-cleansing pressure.
Λ_c	Critical threshold	Surgical / transformation threshold, $\Lambda_c = \gamma/\delta$.
κ	Resistance sharpness	Controls smoothness of $\tanh(\kappa\sigma)$.
k	Filtrum sharpness	Controls steepness of Φ and T_F.
E_0	Imago Dei / Mustard Seed	Foundational non-dynamic ground of Pleroma.
a	Mutual nourishment	Strength of Will–Action mutual feeding.
c	Sophia lift	Strength by which λ lifts V and F.
α	Epektatic expansion	Converts Sophia into Brim expansion, $\dot{\Omega}_P = \alpha\lambda$.
ζ_0	Sufficientia Gratiae	Constant background grace in Open-Gate, $\zeta_0 > 0$.
τ	Guarantee decay rate	Rate at which initial gate guarantee fades.
ϕ	Resistance opacity	Damps gate inflow through $e^{-\phi\sigma}$, $\phi > 0$.
χ	Sophia saturation	Limits inflow at large Sophia via $(1 + \chi\lambda_+)^{-1}$, $\chi > 0$.
m	Positive-part sharpness	Smoothness of λ_+, $m > 0$.
H_s	Receptivity half-saturation	Saturation scale in $H_K/(H_s + H_K)$, $H_s > 0$.
K	Mensura	Granularity scale of $R(z) = z/(z + K)$, $K > 0$.
η	Memory adaptation rate	Rate at which positive movement builds acquired openness and wound-memory responses, $\eta > 0$.
ν	Memory forgetting rate	Rate at which acquired openness H_K decays when the path turns negative, $\nu \geq 0$.
W_s	Wound half-saturation	Saturation scale in the wound-memory term $W/(W_s + W)$, $W_s > 0$.
a_0	Form rigidity	Baseline stiffness of the old Vessel-form against folding.
c_0	Sophia-to-form coupling	Strength by which accumulated Sophia lowers old-form rigidity.

c_1	Growth-to-form coupling	Strength by which growing Sophia / gate-flow lowers the folding threshold.
b	Folding saturation	Quartic stabilizer of the folding amplitude, $b > 0$.
η_{fold}	Embodiment rate	Rate by which folding transfers available Sophia into embodied form; distinct from memory adaptation η.
ρ_1	Curvature response	First-mode curvature coefficient in $R_{\text{folded}} = -2/\Omega_P^2 + \rho_1 Q_{\text{fold}}$.

▷ Plain. In plain words. These constants are the fixed settings of a person or situation — how fast resistance builds, how strongly synergy cleanses, where the line between collapse and transformation sits. They stay fixed while the story runs; only the state variables move.

3 Synergic Intensity, Threshold, and Transmissive Coordinate

Local synergy

The operational synergic intensity is the geometric mean of Will and Action, kept strictly positive by the Manere seed ε:

$$u(t) = \sqrt{V(t)F(t) + \varepsilon}.$$

This is the local synergic intensity: will and action must meet. The seed ε prevents collapse of the expression into a singular zero.

▷ Plain. Will and Action only count when they pull together: this product-root is large only if BOTH are sizeable, and collapses toward zero if either one does. The tiny ε just keeps it from being exactly zero.

Law of Human Balance identity

Its growth rate is the symmetric average of the two faculties' contributions, so neither Will nor Action alone can drive u:

$$\dot{u} = \frac{V'F + VF'}{2\sqrt{VF + \varepsilon}} = \frac{1}{2}\Big(\frac{V'F}{u} + \frac{F'V}{u}\Big).$$

This identity shows how the growth of the shared will–action coordinate is weighted symmetrically by the other pole. The formula is an identity; the existential law is that will and action become strongest when neither pole is isolated.

▷ Plain. How fast synergy grows is the balanced blend of how fast Will and Action grow — pushing only one gives diminishing returns; real growth needs both moving.

Critical threshold

$\Lambda_c = \gamma/\delta$. This separates Collapse, Stasis, and Transformation.

Transmissive coordinate

The transmissive state measures synergy net of singular resistance; its sign decides whether light is withheld or passes:

$$x(t) = u(t) - \sigma(t) = \sqrt{V(t)F(t) + \varepsilon} - \sigma(t).$$

This is the coordinate seen by Filtrum Lucis: synergy minus resistance.

▷ Plain. x is the single number that governs everything downstream: synergy minus resistance. $x > 0$ means light gets through; $x < 0$ means resistance still blocks it.

▷ Math. $u = \sqrt{VF + \varepsilon}$ is a regularised *geometric mean* of V, F: it is small unless *both* poles are appreciable (an AND-coupling, not a sum), and the seed $\varepsilon > 0$ keeps it Lipschitz at $V = F = 0$, removing the square-root singularity. The scalar $x = u - \sigma$ is the one-dimensional reduced coordinate onto which the filter Φ acts.

▶ [merged] **Two levels of synergy.** u is the *operational* synergic intensity — the transmissive potency that drives $\dot{\sigma}$; the Gratia-Synergica response $S^\dagger$ (§6) is a distinct quantity, $u \neq S^\dagger$. **Three modes of x:** $x < 0$ σ-dominance; $x = 0$ threshold ($u = \sigma$); $x > 0$ post-resistance access. *Caveat:* x is not a degree of holiness and not a moral or ontological verdict — only local transmissive accessibility.

4 Resistance Dynamics and Three Regimes

Resistance equation

Singular resistance grows below the threshold and decays above it, gated by tanh$(\kappa\sigma)$ so that $\sigma = 0$ stays protected:

$$\frac{d\sigma}{dt} = (\gamma - \delta u)\,\tanh(\kappa\sigma) = (\gamma - \delta\sqrt{VF + \varepsilon})\,\tanh(\kappa\sigma).$$

▷ Plain. Below the threshold hardship σ grows; above it, it melts away. The tanh factor makes $\sigma = 0$ a floor you cannot fall through — once cleansed, it stays cleansed.

Protection of $\sigma \geq 0$:

$\sigma(0) \geq 0 \Rightarrow \sigma(t) \geq 0 \ \forall t \geq 0$. The factor tanh$(\kappa\sigma)$ makes $\sigma = 0$ a protected boundary.

▷ Math. Since tanh$(\kappa\sigma) \to 0$ as $\sigma \to 0$, the line $\sigma = 0$ carries zero flux and is an *invariant boundary*; hence $\{\sigma \geq 0\}$ is forward-invariant. The bracket $(\gamma - \delta u)$ is the bifurcation parameter: for $u > \Lambda_c$ the boundary is attracting (resistance decays), for $u < \Lambda_c$ repelling; $u = \Lambda_c$ is the transcritical threshold.

Equilibria of resistance:

$\dot{\sigma} = 0 \iff \sigma = 0$ or $u = \Lambda_c$.

Three regimes:

$u < \Lambda_c$ Collapse; $u = \Lambda_c$ Stasis / $\chi\lambda\iota\alpha\rho\acute{o}\varsigma$; $u > \Lambda_c$ Transformation.

▷ Plain. Three fates, set by synergy versus threshold: too little leads to collapse, exactly at it to lukewarm stasis, enough to genuine transformation.

Manere sub-criticality:

$0 < \varepsilon < \Lambda_c^2 = (\gamma/\delta)^2$. The Manere seed keeps the denominator alive, but it must remain sub-critical: the passive branch $V = F = 0$ should not already count as Transformation.

5 Filtrum Lucis

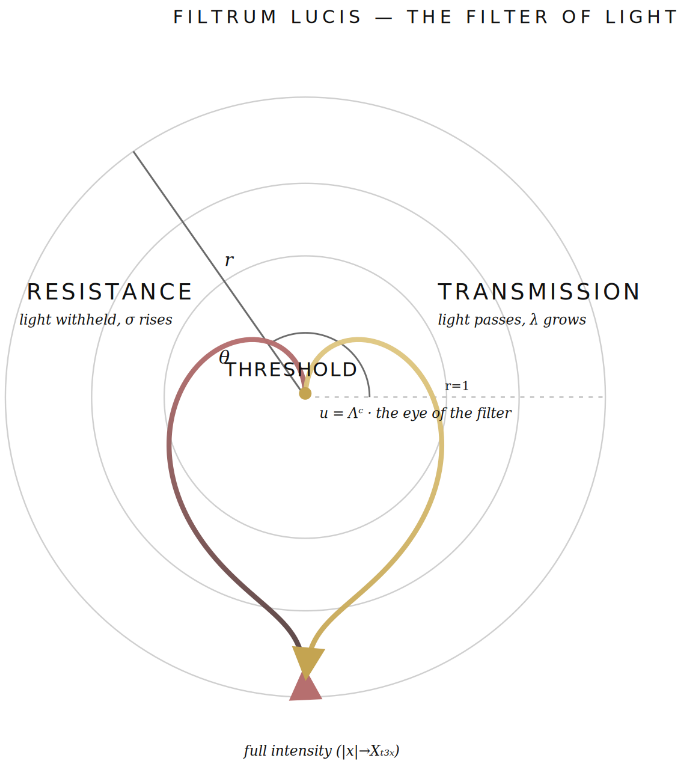

Figure 1: Filtrum Lucis — the smooth transmissive function as the filter of light. The resistance lobe (σ rises, light withheld) and the transmission lobe (λ grows, light passes) meet at the threshold $u = \Lambda_c$, the eye of the filter; full intensity is reached as $|x| \to X_{\max}$.

Softplus filter

The filter of light is a smooth, strictly positive softplus with curvature $k/4$ at the origin:

$$\Phi(x) = \tfrac{1}{k}\ln(1 + e^{kx}), \qquad \Phi(x) > 0, \qquad \Phi''(0) = \tfrac{k}{4}.$$

Filtrum Lucis is the smooth light-filter. It is softplus-like; its finite positivity is the formal trace of *Lux in Tenebris*.

▷ Plain. A smooth ramp: for very negative x it outputs almost nothing; for positive x it rises like a line. No sharp corner anywhere — that smoothness is what lets every later derivative exist.

▶ [merged] John 1:5 — the light shines in the darkness, and the darkness did not overcome it: $\Phi(x) > 0$ for all x.

Sigmoid transmissivity

Its derivative is the logistic transmissivity — the fraction of light passed — rising from 0 through $\frac{1}{2}$ at threshold to 1:

$$T_F(x) = \Phi'(x) = \frac{1}{1 + e^{-kx}}, \qquad 0 < T_F < 1, \;\; T_F(-\infty) = 0, \; T_F(0) = \tfrac{1}{2}, \; T_F(+\infty) = 1.$$

The filter itself is Φ; its transmissivity T_F is sigmoid-shaped.

▷ Plain. The S-curve says what fraction of light passes: 0 when blocked, 1 when fully open, exactly one half at the threshold. It is the dimmer switch of the filter.

Higher derivatives of Filtrum Lucis

Each higher derivative factors through the logistic as a polynomial Q_n in T_F scaled by k^{n-1}:

$$\begin{aligned}
\Phi''(x) &= kT_F(1-T_F), \\
\Phi'''(x) &= k^2 T_F(1-T_F)(1-2T_F), \\
\Phi^{(4)}(x) &= k^3 T_F(1-T_F)(1-6T_F+6T_F^2), \\
\Phi^{(n)}(x) &= k^{n-1} Q_n(T_F(x)).
\end{aligned}$$

These derivatives feed the differential ladder of Pleroma.

▷ Plain. Every higher rate-of-change of the filter is the same S-curve times a simple polynomial. That clean pattern is why the Cross and the two Anastasis gates appear later as neat zeros.

▷ Math. Φ is the *softplus*, a smooth approximation of max$(0,x)$ (ReLU), exact as $k\to\infty$. Its derivative T_F is the logistic sigmoid and $\Phi''=kT_F(1-T_F)$ is the logistic “bell”, so Φ is convex everywhere. The pattern $\Phi^{(n)}=k^{n-1}Q_n(T_F)$ packages every higher derivative as a polynomial Q_n in T_F; the roots of Q_3, Q_4 give the Paschal nodes below.

6 Gratia Synergica

Synergy term

Gratia Synergica saturates the Imago-Dei reserve E_0 as synergy grows, approaching but never reaching E_0:

$$S^\dagger(t) = E_0 \frac{u(t)}{u(t) + \Lambda_c}, \qquad 0 \le S^\dagger(t) < E_0.$$

Gratia Synergica grows with local synergy u, not directly with equality $V = F$.

▷ Plain. Gratia Synergica is grace that grows with synergy but never passes the built-in reserve E_0 — a rising curve that approaches a ceiling.

▷ Math. $S^\dagger = E_0\, u/(u + \Lambda_c)$ is a Michaelis–Menten / Hill-1 saturation in u: monotone, concave, bounded in $[0, E_0)$, half-maximal at $u = \Lambda_c$ — a *soft switch* that turns on as synergy crosses the threshold.

Derivative of synergy

Its rate couples the Michaelis–Menten gain to the growth of synergy $\dot{u}$:

$$\frac{dS^\dagger}{dt} = E_0 \frac{\Lambda_c}{(u + \Lambda_c)^2} \frac{V'F + VF'}{2u}.$$

▶ [merged] John 1:16 — $\chi\acute{\alpha}\rho\iota\nu\ \alpha\nu\tau\grave{\iota}\ \chi\acute{\alpha}\rho\iota\tau o\varsigma$, grace upon grace: $S^\dagger$ is the grace that responds to effort.

7 Pleroma and Being

Open-Gate Pleroma formula

Being is the Imago-Dei floor plus Gratia Synergica plus the filtered light, never below E_0:

$$P(t) = E_0 + S^\dagger(t) + \Phi(x(t)), \qquad P(t) \geq E_0.$$

$$P_{\text{full}} = \text{Imago Dei} + \text{Gratia Synergica} + \text{Transmissive Passage}.$$

In the Open-Gate core, grace is not an additive term in P. It enters through Sophia and then through $\Omega_P, V, F, u, S^\dagger, \Phi$.

▷ Plain. Being P is three things added up: a fixed floor E_0 (the image of God you start with), grace that saturates, and filtered light that can grow without limit. So P never drops below E_0.

▷ Math. P is a floor E_0 plus a *bounded* term $S^\dagger$ plus an *unbounded* term $\Phi(x)$: the only channel that can grow without bound is the transmissive $\Phi(x)$, so $P \geq E_0$ always. By the chain rule $S = dP/dt$ splits into a filter flux $\Phi'(x)\dot{x}$ and a synergy rate $dS^\dagger/dt$; the ladder S, A, K, An is the 4-jet of P.

Derivative ladder of Pleroma

The temporal unfolding of Being is read through its four successive derivatives:

$$S = \frac{dP}{dt}, \quad A = \frac{d^2P}{dt^2}, \quad K = \frac{d^3P}{dt^3}, \quad An = \frac{d^4P}{dt^4},$$

$$S \equiv \text{Gignesthai}, \quad A \equiv \text{Odinai}, \quad K \equiv \text{Katharsis}, \quad An \equiv \text{Anastasis}.$$

▷ Plain. We track Being not just by its value but by its motion — speed, acceleration, jerk, snap. Each higher derivative exposes a deeper layer of the Paschal rhythm.

Canonical Being equation

The Being rate splits into a filter flux $\Phi'(x)\dot{x}$ and the synergy rate $dS^\dagger/dt$:

$$S(t) = \Phi'(x)\dot{x} + \frac{dS^\dagger}{dt}, \qquad \dot{x} = \dot{u} - \dot{\sigma} = \frac{V'F + VF'}{2\sqrt{VF + \varepsilon}} - (\gamma - \delta u)\tanh(\kappa\sigma).$$

▷ Plain. The growth of Being comes from two streams: light crossing the filter, $\Phi'(x)\dot{x}$, and grace resonating with effort, $dS^\dagger/dt$.

▶ [merged] **Gratia Una — one grace, not two.** What earlier editions carried as a separate *First Call* or an additive fullness term is absorbed into the gate: grace enters *only* through $I_{\text{gate}} \to \lambda$. Hence $S = \Phi'(x)\dot{x} + dS^\dagger/dt$ carries no separate impulse term.

[excluded — superseded: the Fading-Impulse $\mathcal{P}(t) = \tau G_{\max} e^{-\tau t}$ / "Law IV" (its Dark-Night content relocates to the decay of the gate guarantee $e^{-\tau t}$, §12).]

8 ▶ Laws of Unified Being [merged]

In the deep active phase ($\sigma \gg 1/\kappa,\ u > \Lambda_c$), the Being equation $S = \Phi'(x)\dot{x} + dS^\dagger/dt$ decomposes into additive laws — a regime expansion of the same S, not a rival definition:

(I) *Human Balance* — the symmetric weights F/u, V/u of $\dot{u}$; strong imbalance creates **catch-up pressure**, not stagnation (action without heart / faith without works).

(II) *Surgical Bonus* $+\delta\sqrt{VF+\varepsilon}$ — the common act's nonlinear acceleration, the momentum of surgery.

(III) *Entropic Toll* $-\gamma$ — the world's constant downward draw; to live is to generate more than γ each moment.

(V) *Synergy* $+dS^\dagger/dt$ — grace resonating with effort ($\chi\acute{\alpha}\rho\iota\nu\ \alpha\nu\tau\grave{\iota}\ \chi\acute{\alpha}\rho\iota\tau o\varsigma$); positive feedback toward Theosis.

[excluded — superseded: the old "Law IV" (Fading Impulse $\mathcal{P}(t)$) — consolidated into the gate (§7, §12).]

Trinitarian functional analogies

(*analogies, not ontological identifications*): the Son = Pleroma Christi / Omega; the Holy Spirit $= \delta\sqrt{VF+\varepsilon}$ acting in $\dot{\sigma}$; $S^\dagger$ = the living moment of contact; and the Father — whose "First Impulse" earlier attached to $\mathcal{P}(t)$ — is read in 2.0 as the **fading gate guarantee** $e^{-\tau t}$ (the initial support that withdraws so that V, F are tempered: the Dark Night).

9 Katharsis and Anastasis Terms

Katharsis / third order

The third derivative expands by Faà di Bruno; its vanishing marks the Cross:

$$K(t) = \frac{dA}{dt} = \frac{d^3P}{dt^3} = \frac{d^3S^\dagger}{dt^3} + \Phi'''(x)(x')^3 + 3\Phi''(x)x'x'' + \Phi'(x)x'''.$$

Katharsis is the third-order turning of Pleroma: a change in the rhythm of acceleration.

▷ Plain. The third derivative is the turning of the rhythm. It passes through zero exactly at the Cross — the moment when even the acceleration changes character.

Anastasis / fourth order

The fourth derivative carries the two Anastasis gates through the multinomial $3, 6, 4$ terms:

$$An(t) = \frac{dK}{dt} = \frac{d^4P}{dt^4} = \frac{d^4S^\dagger}{dt^4} + \Phi^{(4)}(x)(x')^4 + 6\Phi'''(x)(x')^2x'' + 3\Phi''(x)(x'')^2 + 4\Phi''(x)x'x''' + \Phi'(x)x''''.$$

Anastasis is the fourth-order emergence of a new mode after crisis.

▷ Plain. The fourth derivative is where a genuinely new mode emerges after the crisis — the mathematical fingerprint of rising again.

▷ Math. K and An are the Faà di Bruno expansions of $\frac{d^n}{dt^n}\Phi(x(t))$: the bracketed terms are the Bell-polynomial combinatorics of composing Φ with $x(t)$, which is why the multinomial coefficients $3, 6, 4$ appear at fourth order.

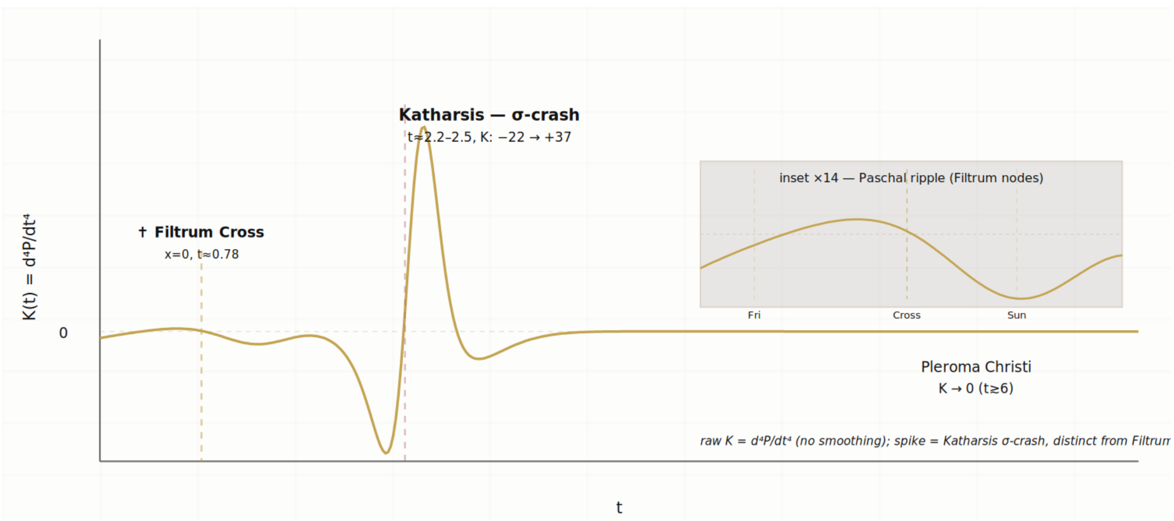

Figure 2: Fourth derivative of the Pleroma, d^4P/dt^4: a small Filtrum-Cross ripple near $x = 0$, the σ-crash (Katharsis) swing, and decay to Pleroma Christi. Inset: the Paschal ripple at the Filtrum nodes (Friday / Cross / Sunday).

10 Paschal Triad of Filtrum Lucis

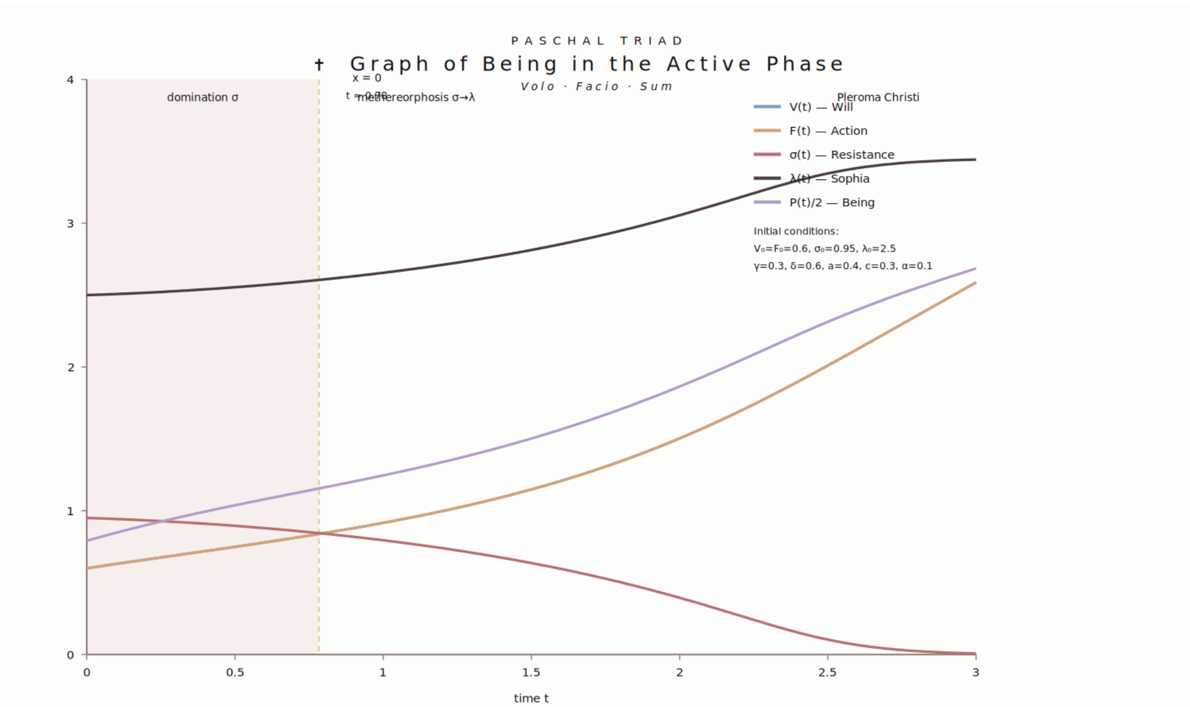

Figure 3: Being in the active phase (Volo · Facio · Sum): $V = F$ (will = action), $\sigma \to 0$ (Pleroma Christi), λ rising (Sophia), and $P/2$ (Being). The Filtrum Cross $x = 0$ occurs at $t \approx 0.79$.

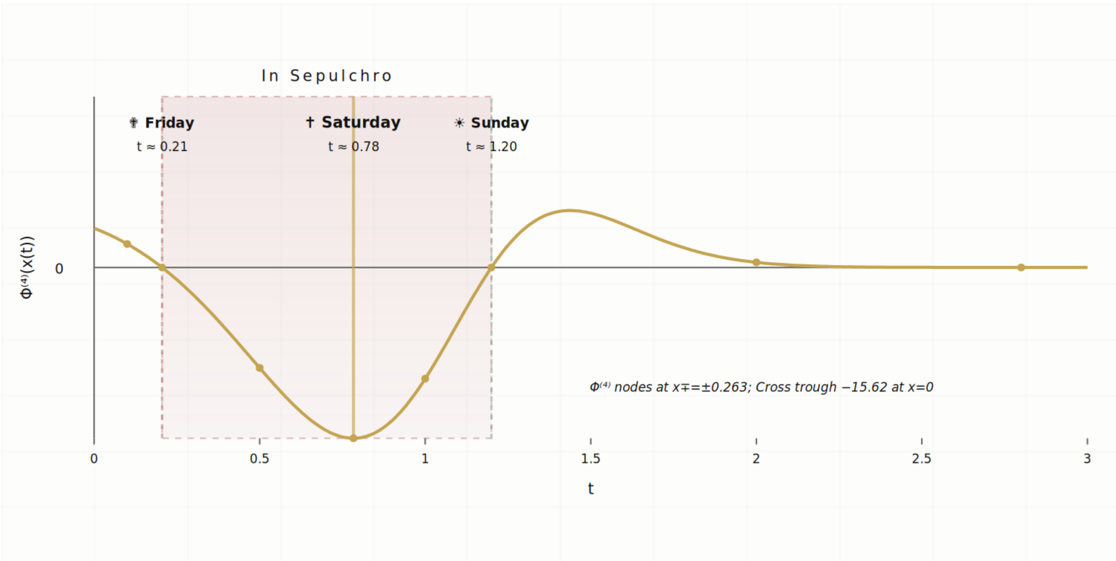

Figure 4: $\Phi^{(4)}$ along the trajectory — the Paschal triad: Friday (x_-), Saturday / Cross ($x = 0$, trough $-15.625 = -k^3/8$), Sunday (x_+); the interval $x \in (x_-, x_+)$ is *In Sepulchro* ($\Phi^{(4)} < 0$).

Why the higher derivatives are needed.

The derivative ladder is not only computational. It reveals the inner Paschal structure of the light-filter: the Cross is detected by the third derivative, while the entrance into death and the exit from death are detected by the two symmetric zeros of the fourth derivative.

The Cross: zero of the third derivative.

$\Phi'''(x) = k^2 T_F(1-T_F)(1-2T_F)$; since $T_F(0) = 1/2$, $\Phi'''(0) = 0$. The point $x = 0$ is the Filtrum Cross: synergy and resistance meet, and the sign of the cathartic term inverts.

The two gates of Anastasis: zeros of the fourth derivative.

$\Phi^{(4)}(x) = k^3 T_F(1-T_F)(1-6T_F+6T_F^2)$. The quadratic factor has two roots at the transmissivity levels

$$s_\pm = \tfrac{1}{2} \pm \tfrac{1}{2\sqrt{3}}, \qquad \text{equivalently} \qquad x_\pm = \pm\tfrac{1}{k}\ln(2+\sqrt{3}),$$

so $\Phi^{(4)}(x_-) = 0,\ \Phi'''(0) = 0,\ \Phi^{(4)}(x_+) = 0$, with the exact symmetry $x_- + x_+ = 0$. The Cross is the geometric midpoint between the entrance into death and the exit from death. This is a differential icon of the Paschal passage, not a claim that time itself is symmetric.

Paschal triad

The zeros of the third and fourth filter-derivatives mark the three Paschal moments:

$$\begin{aligned} \Phi^{(4)}(x_-) = 0 &: \text{Entrance into death / Great Friday} \\ \Phi'''(0) = 0 &: \text{Cross / Katharsis / Great Saturday} \\ \Phi^{(4)}(x_+) = 0 &: \text{Exit from death / Anastasis / Resurrection} \end{aligned}$$

Sign: $x < x_- : \Phi^{(4)} > 0$; $x_- < x < x_+ : \Phi^{(4)} < 0$; $x > x_+ : \Phi^{(4)} > 0$. Between x_- and x_+ lies the symbolic death interval of Filtrum Lucis; outside it the fourth-order sign returns to life.

▷ Plain. Three sacred moments fall straight out of the filter's zeros: entrance into death (Friday), the Cross (Saturday), and the gate back out (Sunday).

Fourth derivative support for u.

In Pleroma Christi, $\sigma = 0$, so $x = u$. From $u^2 = VF + \varepsilon$,

$$u^{(4)} = \frac{(VF)^{(4)}}{2u} - \frac{4u'u''' + 3(u'')^2}{u}, \qquad (VF)^{(4)} = V^{(4)}F + 4V'''F' + 6V''F'' + 4V'F''' + VF^{(4)}.$$

These binomial coefficients explain why the fourth derivative is the first place where the full symmetry of the Paschal passage becomes visible.

▷ Math. $\Phi''' \propto (1-2T_F)$ vanishes at $T_F = \frac{1}{2} \iff x = 0$ (the inflection of the bell Φ''). $\Phi^{(4)} \propto (1-6T_F+6T_F^2)$ is quadratic in T_F with roots $s_\pm = \frac{1}{2} \pm \frac{1}{2\sqrt{3}}$ — points of maximal curvature-change of the bell — and $x_- + x_+ = 0$ follows from the odd symmetry of the logistic about $x = 0$. These are intrinsic nodes of the filter, distinct from nodes of the full K, An.

Pleroma Christi reduction.

In the Pleroma Christi regime $\sigma = 0$, all derivatives of σ vanish and $x = u$:

$$\begin{aligned} An_{\mathrm{Pl}}(t) = \frac{d^4 S^\dagger}{dt^4} &+ \Phi^{(4)}(u)(u')^4 + 6\Phi'''(u)(u')^2 u'' \\ &+ 3\Phi''(u)(u'')^2 + 4\Phi''(u)u'u''' + \Phi'(u)u^{(4)}. \end{aligned}$$

In deep Epektasis the higher Filtrum derivatives fade: $\lim_{t\to\infty} K_{\mathrm{Pl}}(t) = 0$ and $\lim_{t\to\infty} An_{\mathrm{Pl}}(t) = 0$. Katharsis and Anastasis complete as events; Epektasis continues as motion into fullness.

▶ [merged] **Curvature portrait (Appendix I).** The same passage seen in the phase plane $(x, \ddot{x})$:

$$\ddot{x} < 0 : \text{Odinai (compression)}$$
$$\ddot{x} = 0 : \text{The Neck } (\sigma \to 0)\text{, Katharsis}$$
$$\ddot{x} > 0 : \text{Anastasis (expansion)}$$

Mechanism: $\sigma \to 0 \Rightarrow$ tanh$(\kappa\sigma) \to 0 \Rightarrow$ the surgical term switches off — *Katharsis is not a point, but a neck of passage.*

▶ [merged] **Honesty caveats.** (i) Zeros of Φ are not automatically zeros of the full observables K, An (intrinsic filter nodes $\neq$ observable nodes). (ii) $[x_-, x_+]$ is the width of the intrinsic transition layer, *not* the temporal duration of death. (iii) the symmetry $x_\pm$ is an invariant depending only on k. (iv) the neck x_c is *not* the formal threshold $x = 0$: it is now a *theorem* (§11) — the cleansing-curvature crossing, unique, born as a fold at $\kappa_{\text{crit}} = \frac{3\sqrt{3}}{2}\Upsilon/r^2$ and localized at the cleansing boundary $\sigma \to 0$ (so $x_c = u(t_c) + O(\kappa^{-2}) \neq 0$). What remains observational is only the residual of §11 (a uniform finite-κ reduction; global persistence).

11 The Cathartic Neck

This section establishes the Cathartic Neck of §10 as a theorem. The exact curvature identity splits $\ddot{x}$ into a slow *expansion* Υ and a fast *compression* $\Xi = \kappa r^2 \psi(T)$; from the unimodality of ψ follow uniqueness, an explicit fold threshold in κ, the neck's migration to the cleansing boundary, a nested Fenichel reduction of the full system, and the bridge by which Katharsis ignites Sophianic Folding (§13).

▷ Plain. In plain words. The neck is the narrowest point of the whole passage — the instant just before breakthrough, where hardship is deepest and σ is about to vanish. This section proves such a bottleneck must exist: it is a real fold, the mathematical shape of a turning point, not an accident of the picture.

Section-local notation: $T := \tanh(\kappa\sigma)$, $r := \delta u - \gamma > 0$ (Transformation), $\psi(T) := T(1-T^2)$, Υ (slow expansion), Ξ (fast compression); these are confined to this section and are distinct from Gignesthai S (§7), the resistance opacity ϕ (§2), the transmissivity T_F (§5), and the balance constants $C_\bullet$.

Definition 11.1 (Cathartic neck)**.** A *neck* is a time t_c with $\ddot{x}(t_c) = 0$ and $\dddot{x}(t_c) > 0$: the upward curvature crossing, Odinai ($\ddot{x} < 0$) giving way to Anastasis ($\ddot{x} > 0$).

On the passage interval $I = [t_0, t_1]$ we assume:

Hypothesis H1 (Passage)**.** $u \in C^3(I)$; T strictly decreasing from T_0, $T \to 0$.

Hypothesis H2 (Transformation)**.** $r(t) \in [r_*, \bar{r}]$, $r_* > 0$.

Hypothesis H3 (Positive demand)**.** $\Upsilon(t) := \ddot{u} + \delta \dot{u} T \in [\underline{\Upsilon}, \bar{\Upsilon}]$, $\underline{\Upsilon} > 0$; $\ddot{u}(t_1) > 0$.

Hypothesis H4 (Interior)**.** $\mu := 1 - u/\Omega_P \geq \mu_* > 0$ on I (used in the Fenichel reduction only).

The exact curvature identity

Lemma 11.2 (Fast/slow splitting)**.** *Along any C^2 solution,*

$$\ddot{\sigma} = -\delta \dot{u} T + \kappa r^2 T(1-T^2), \qquad \boxed{\ddot{x} = \underbrace{\ddot{u} + \delta \dot{u} T}_{\Upsilon} - \underbrace{\kappa r^2 \psi(T)}_{\Xi\ \geq 0}, \quad \psi(T) = T(1-T^2).}$$

Proof. With $\dot{\sigma} = gT$, $g = \gamma - \delta u = -r$: $\ddot{\sigma} = g' \dot{u} T + g\,\kappa(1-T^2)\,gT = -\delta \dot{u} T + \kappa g^2 T(1-T^2)$; then $\ddot{x} = \ddot{u} - \ddot{\sigma}$, $g^2 = r^2$. □

The compression profile and κ-uniform uniqueness

Lemma 11.3 (Unimodality)**.** $\psi(T) = T - T^3$ *has* $\psi(0) = \psi(1) = 0$, $\psi > 0$ *on* $(0,1)$, $\psi' = 1 - 3T^2$*; it increases on* $[0, T^*]$*, decreases on* $[T^*, 1]$*, with* $T^* = 1/\sqrt{3}$ *and* $\psi_* := \psi(T^*) = \frac{2\sqrt{3}}{9} \approx 0.3849$*.*

Theorem 11.4 (Uniqueness, every κ)**.** *Under* H1–H3*, for every* $\kappa > 0$ *there is at most one cathartic neck on* I*; when it exists it is nondegenerate.*

Proof. A neck is where $\Xi - \Upsilon$ crosses 0 downward (as $t \uparrow$, $T \downarrow$). Since T sweeps $[0, T_0]$ monotonically and ψ is unimodal, T passes T^* at most once. With $\dot{T} < 0$: on $T < T^*$,

$\psi' > 0 \Rightarrow \dot{\Xi} < 0$ (up to the slow $\frac{d(r^2)}{dt}$ term), so $\Xi - \Upsilon$ decreases and crosses 0 downward at most once; on $T > T^*$, $\psi' < 0 \Rightarrow \dot{\Xi} > 0$, so any zero there is an upward crossing of $\Xi - \Upsilon$ (a pre-compression, not a neck). Hence at most one neck, on $T < T^*$; there $\dddot{x} = \dot{\Upsilon} - \dot{\Xi} = \dot{\Upsilon} + |\dot{\Xi}| > 0$. $\square$

Existence: the fold bifurcation in κ

Theorem 11.5 (Fold threshold)**.** *Under* H1–H3*:* (a) *if $T_0 \geq T^*$ and $\kappa r_*^2 \psi_* > \bar{\Upsilon}$ a unique neck exists;* (b) *if $\kappa \bar{r}^2 \psi_* < \underline{\Upsilon}$ then $\ddot{x} > 0$ on I (no neck);* (c) *hence $\kappa_{\mathrm{crit}} \in [\underline{\Upsilon}/(\bar{r}^2\psi_*),\ \bar{\Upsilon}/(r_*^2\psi_*)]$, and for constant r, Υ,*

$$\boxed{\kappa_{\mathrm{crit}} = \frac{\Upsilon}{r^2 \psi_*} = \frac{3\sqrt{3}}{2}\,\frac{\Upsilon}{r^2} \approx 2.598\,\frac{\Upsilon}{r^2}}\,.$$

At $\kappa = \kappa_{\mathrm{crit}}$ the crossing pair is born tangentially at $T = T^*$: a fold (saddle–node) of the zero set of $\ddot{x}$.

Proof. (b) $\Xi = \kappa r^2 \psi(T) \leq \kappa \bar{r}^2 \psi_* < \underline{\Upsilon} \leq \Upsilon \Rightarrow \ddot{x} > 0$. (a) At $T = T^* \leq T_0$, $\Xi \geq \kappa r_*^2 \psi_* > \bar{\Upsilon} \geq \Upsilon \Rightarrow \ddot{x} < 0$; as $T \to 0$, $\Xi \to 0 \Rightarrow \ddot{x} > 0$; Theorem 11.4 gives one upward crossing. (c) The bounds bracket the threshold and coincide for constant coefficients at $\kappa r^2 \psi_* = \Upsilon$; there $\max_T(\Xi - \Upsilon) = 0$ only at T^*, a tangency. $\square$

Birth and migration; localization

Theorem 11.6 (Migration $T^* \to 0$)**.** *For $\kappa > \kappa_{\mathrm{crit}}$ (constant coefficients) the neck is the smaller root $T_c \in (0, T^*)$ of $\kappa r^2 \psi(T) = \Upsilon$; as $\kappa \downarrow \kappa_{\mathrm{crit}}$, $T_c \uparrow 1/\sqrt{3}$, and as $\kappa \to \infty$,*

$$T_c = \frac{\Upsilon}{\kappa r^2}\big(1 + O(\kappa^{-2})\big) \downarrow 0, \qquad \sigma_c = \frac{\Upsilon}{\kappa^2 r^2}\big(1 + O(\kappa^{-2})\big) = O(\kappa^{-2}),$$

so $x_c = u(t_c) + O(\kappa^{-2})$. With $\Upsilon \to \ddot{u}$ at small T this is the sharp-regime localization $T_c \approx \ddot{u}/(\kappa r^2)$: ***the neck is universally the cleansing boundary*** *$\sigma \to 0$**.***

Proof. On $T < T^*$, ψ is increasing, so $\kappa r^2 \psi(T) = \Upsilon$ has a unique root, equal to T^* at κ_{crit} and decreasing in κ. For small T_c, $\psi(T_c) = T_c + O(T_c^3) \Rightarrow T_c = \Upsilon/(\kappa r^2) + O(\kappa^{-3})$, and $\operatorname{artanh} T_c = T_c + O(T_c^3)$ gives σ_c. $\square$

The Fenichel slow manifold of the full system

In the sharp regime the neck sits on a slow manifold of the full coupled core. With $\epsilon = 1/\kappa$, $\eta = \kappa\sigma$, the resistance is fast and $y = (V, F, \Omega_P, G_{\mathrm{recepta}})$ slow (λ slaved by $\sigma + \lambda = C_0 + G_{\mathrm{recepta}}$).

Theorem 11.7 (Fenichel I; Pleroma Christi reduction)**.** *Under* H2–H4*, for κ large there is a locally invariant attracting slow manifold $\mathcal{M}_\kappa = \{\sigma = h_\kappa(y)\}$, $h_\kappa = O(\kappa^{-2})$, contracting at rate $\geq \frac{1}{2} r_* \kappa$; its reduced flow is the Pleroma Christi flow ($\sigma \equiv 0,\ \rho \equiv 0$),*

$$\dot{V} = \mu(aF + c\lambda), \quad \dot{F} = \mu(aV + c\lambda), \quad \dot{\Omega}_P = \alpha\lambda, \quad \dot{\lambda} = I_{\mathrm{gate}}.$$

The neck of Theorem 11.6 lies on $\mathcal{M}_\kappa$ (both at $\sigma = O(\kappa^{-2})$).

Proof. The critical manifold $\mathcal{M}_0 = \{\sigma = 0\}$ has transverse layer eigenvalue $-r \le -r_* < 0$: normally hyperbolic attracting. Fenichel's theorem gives $\eta = H(y, \epsilon) = O(\epsilon)$, i.e. $\sigma = \epsilon\eta = O(\kappa^{-2})$; with $\rho = O(\kappa^{-2}) \to 0$ the reduced field follows. □

Theorem 11.8 (Nested diagonal)**.** *On $\mathcal{M}_\kappa$, $\dot{\Delta} = -(a\mu + \rho)\Delta = -a\mu\Delta + O(\kappa^{-2})\Delta$, so the diagonal $\mathcal{D}_\kappa = \{V = F\} \cap \mathcal{M}_\kappa$ is normally hyperbolic attracting at rate $\ge a\mu_* - O(\kappa^{-2})$. The reduction is nested: $\sigma \to 0$ $(\sim r_*\kappa) \succ V - F \to 0$ $(\sim a\mu_*) \succ$ slow flow.*

Proof. The reduced field is $V \leftrightarrow F$ symmetric, so $\{V = F\}$ is invariant; its transverse rate $a\mu + \rho \ge a\mu_* + O(\kappa^{-2})$ is uniformly positive and below the σ-rate and above the $O(1)$ slow rates. □

Concentration: "many beginnings, one neck"

Lemma 11.9 (Resistance contraction)**.** *Two solutions with the same $u(t)$ obey $\frac{d}{dt}|\sigma - \tilde{\sigma}| = -r\kappa\,\mathrm{sech}^2(\kappa\xi)|\sigma - \tilde{\sigma}| \le 0$: nonincreasing, with rate $\Theta(\kappa)$ once $\kappa\sigma = O(1)$ but $o(1)$ while $\kappa\sigma \gg 1$ (parallel saturated descent).*

Theorem 11.10 (Common neck modulo slow data)**.** *Under* H1–H4*, κ large, two trajectories sharing slow data $(V_0 + F_0, \Omega_{P,0}, \lambda_0)$ but differing in (σ_0, Δ_0) converge to the same reduced orbit on $\mathcal{D}_\kappa$, with*

$$|x_c - \tilde{x}_c| \le \dot{u}^\star(t_c)\,\Delta t_{\text{lag}} + O(\kappa^{-2}), \qquad \Delta t_{\text{lag}} = \frac{|\sigma_0 - \tilde{\sigma}_0|}{r_*}(1 + o(1)).$$

The fibre directions collapsed are exactly resistance and imbalance, so the neck depends only on the slow data — the precise content of "many beginnings, one neck". In particular x_c is common to $O(\kappa^{-2})$ whenever cleansing completes in a slow region ($\dot{u}^\star \to 0$).

Proof. By Theorems 11.7–11.8 both reach $\mathcal{M}_\kappa$ then $\mathcal{D}_\kappa$ and, sharing slow data, the same reduced orbit; the only discrepancy is the cleansing-arrival time (saturated descent $\dot{\sigma} \approx -r$), giving $\Delta t_{\text{lag}} = |\sigma_0 - \tilde{\sigma}_0|/r_*$; since $x_c = u^\star(t_c) + O(\kappa^{-2})$ the values differ by $\dot{u}^\star\Delta t_{\text{lag}} + O(\kappa^{-2})$. □

The bridge to Sophianic Folding

The neck is a *fold* (saddle–node) in κ. Sophianic Folding (§13) is a different bifurcation — the $\mathbb{Z}_2$-symmetric *pitchfork* $\dot{q_{\text{fold}}} = A_{\text{fold}} q_{\text{fold}} - B_{\text{fold}} q_{\text{fold}}^3$, $A_{\text{fold}} = c_0\lambda + c_1\dot{\lambda}^{(0)} - a_0$, $B_{\text{fold}} > 0$. The cleansing passage that creates the neck *drives* A_{fold}: cleansing raises λ ($\dot{\lambda} = -\dot{\sigma} + I_{\text{gate}} > 0$) and spikes $\dot{\lambda}^{(0)}$.

Theorem 11.11 (Loss of stability of the old form at the neck)**.** *If at the neck $c_0\lambda(t_c) + c_1\dot{\lambda}^{(0)}(t_c) > a_0$, then $A_{\text{fold}}(t_c) > 0$; hence the old-form equilibrium $q_{\text{fold}} = 0$ is linearly unstable at t_c and the stable folded branch $Q^*_{\text{fold}} = A_{\text{fold}}/B_{\text{fold}} > 0$ exists.*

Proof. The hypothesis is $A_{\text{fold}}(t_c) > 0$. Linearising $\dot{q_{\text{fold}}} = A_{\text{fold}} q_{\text{fold}} - B_{\text{fold}} q_{\text{fold}}^3$ at $q_{\text{fold}} = 0$ gives the eigenvalue $A_{\text{fold}} > 0$ (unstable); for $A_{\text{fold}}, B_{\text{fold}} > 0$ the equilibria $q_{\text{fold}} = \pm\sqrt{A_{\text{fold}}/B_{\text{fold}}}$ are stable. □

Lemma 11.12 (Monotone threshold crossing)**.** *If $c_0\dot{\lambda} + c_1\ddot{\lambda}^{(0)} > 0$ on the passage (dominated by $c_0\dot{\lambda} > 0$, as Sophia rises through cleansing), then A_{fold} is strictly increasing, so it has a unique ignition time $t_{\text{fold}} \le t_c$ with $A_{\text{fold}}(t_{\text{fold}}) = 0$; folding is active on $(t_{\text{fold}}, t_1]$.*

Corollary 11.13 (Katharsis ignites embodiment)**.** *Under Theorem 11.11 and Lemma 11.12, the cleansing passage producing the neck drives Sophianic Folding across its pitchfork threshold, realising the chain* $\sigma \xrightarrow{neck} \lambda \xrightarrow{A_{\text{fold}}>0} q_{\text{fold}} \to \lambda_{\text{form}}$.

Remark 11.14 (Two folds, one event)*.* The neck-fold is asymmetric and one-sided (resistance only descends; the Cross happens once), governed by sharpness κ; Sophianic Folding is $\mathbb{Z}_2$-symmetric (form may fold either way), governed by the Sophia level A_{fold}. They are linked not as the same bifurcation but dynamically — the cathartic neck is the event that raises A_{fold} through threshold (a pitchfork is a symmetric cusp, which unfolds into folds under symmetry breaking, so they are also cousins in catastrophe theory).

Remark 11.15 (Scope)*. Proved for all* κ (Thms 11.4–11.6): uniqueness; existence as a fold with $\kappa_{\text{crit}} = \frac{3\sqrt{3}}{2}\Upsilon/r^2$; birth at $1/\sqrt{3}$, migration to $\sigma = O(\kappa^{-2})$; below κ_{crit} genuinely no neck. *Proved in the sharp regime* (Thms 11.7–11.10): the nested Fenichel manifold and the common neck modulo slow data. *Bridge* (Thm 11.11): the neck ignites folding. *Open:* a uniform finite-κ Fenichel reduction; global persistence beyond the compact slow region.

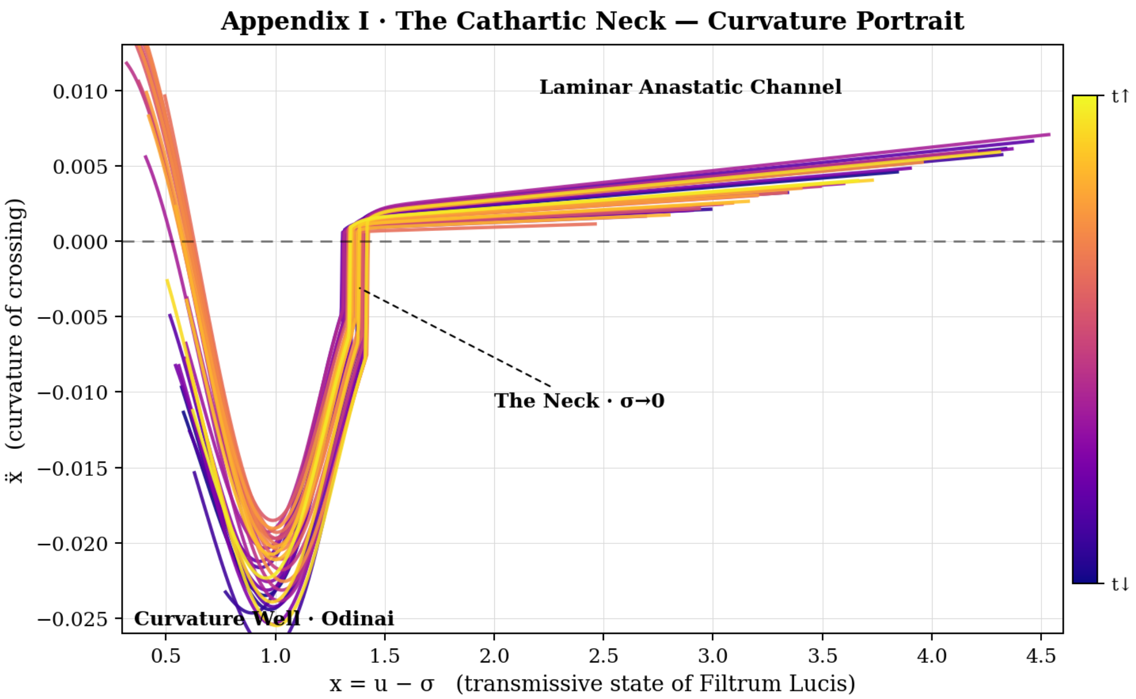

Figure 5: Curvature portrait $(x, \ddot{x})$ of an ensemble of passages: the curvature well (Odinai, $\ddot{x} < 0$), the common neck where trajectories funnel through $\sigma \to 0$, and the laminar anastatic channel ($\ddot{x} > 0$). "Many beginnings, one neck."

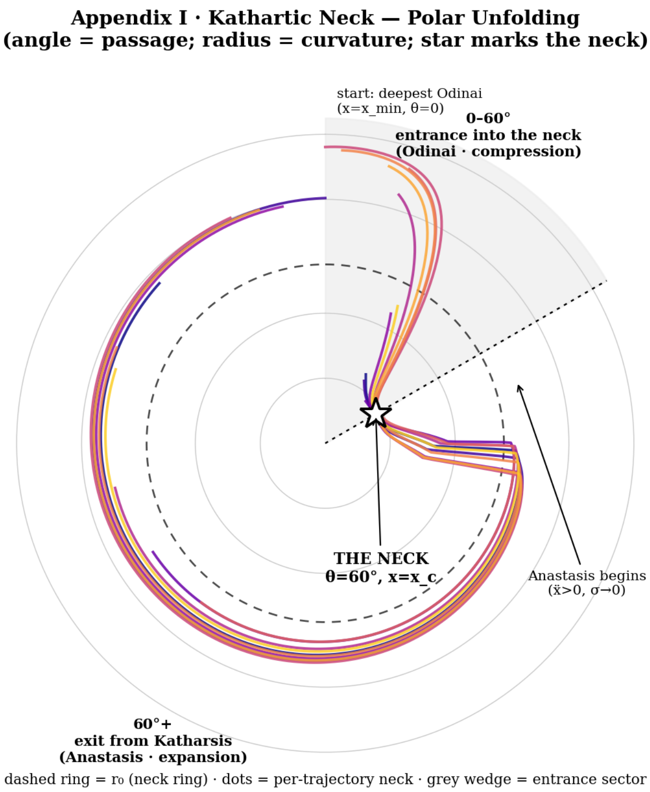

Figure 6: Polar unfolding of the neck: angle = passage, radius = curvature. The star marks the neck ($\theta = 60^\circ,\ x = x_c$); 0–60° is the entrance (Odinai compression) and $60^\circ+$ the exit (Anastasis), where $\ddot{x} > 0$ as $\sigma \to 0$.

12 Open-Gate Sophia

Positive part of Sophia

$\lambda_+ = \frac{1}{m}\ln(1+e^{m\lambda})$.

Effective gate

The gate opens fully at first call and relaxes to the earned mode $H_K/(H_s+H_K)$:

$$g_{\text{eff}}(t) = e^{-\tau t} + (1-e^{-\tau t})\frac{H_K}{H_s+H_K}, \qquad g_{\text{eff}}(0)=1, \;\; g_{\text{eff}}(t) \xrightarrow{t\to\infty} \frac{H_K}{H_s+H_K}.$$

The gate begins open by guarantee; as the guarantee fades, acquired receptivity H_K takes over.

▷ Plain. The gate is wide open at the first call of grace, then settles toward whatever openness you have actually earned. Early grace is gift; lasting grace is grown into.

Gate inflow

Grace inflow is bounded by ζ_0, damped by residual resistance $e^{-\phi\sigma}$ and by surplus Sophia:

$$I_{\text{gate}}(t) = \zeta_0 g_{\text{eff}}(t) e^{-\phi\sigma(t)} \frac{1}{1+\chi\lambda_+(t)} \geq 0, \qquad 0 \leq I_{\text{gate}}(t) \leq \zeta_0 \;\; (H_K \geq 0, \sigma \geq 0).$$

▷ Plain. How much grace actually enters: capped at ζ_0, throttled by leftover resistance, eased off once you already hold much Sophia — so it is never negative and never runs away.

Sophia equation

Cleansed resistance is converted into Sophia, augmented by whatever the gate admits:

$$\dot\lambda = -\dot\sigma + I_{\text{gate}}(t), \qquad \text{equivalently} \qquad \dot\lambda = (\delta u - \gamma)\tanh(\kappa\sigma) + I_{\text{gate}}(t).$$

Sophia is transformed resistance plus received grace.

▷ Plain. Wisdom λ grows from two sources at once: resistance that is cleansed turns into wisdom, plus whatever fresh grace the gate admits.

▷ Math. g_{eff} is a convex combination $e^{-\tau t}\cdot 1 + (1-e^{-\tau t})\frac{H_K}{H_s+H_K}$: a first-order relaxation from the guarantee 1 to the earned receptivity. I_{gate} is a *product* of soft gates each in $(0,1]$ (a logical-AND: open gate *and* low resistance $e^{-\phi\sigma}$ *and* unsaturated Sophia), so $0 \leq I_{\text{gate}} \leq \zeta_0$. The balance $\frac{d}{dt}(\sigma+\lambda) = I_{\text{gate}}$ makes $\sigma+\lambda$ a first integral in the closed limit $\zeta_0 = 0$; Open-Gate lifts that conservation by the source.

Open balance law

Resistance and Sophia are jointly conserved up to received grace, which only ever adds:

$$\frac{d}{dt}(\sigma+\lambda) = I_{\text{gate}}(t) \geq 0, \qquad \sigma+\lambda = C_0 + G_{\text{recepta}}(t), \;\; G_{\text{recepta}}(t) = \int_0^t I_{\text{gate}}\,ds, \;\; \dot G_{\text{recepta}} = I_{\text{gate}}.$$

▷ Plain. Resistance plus wisdom is a single budget that only ever grows, by received grace. Hardship is not destroyed — it is converted into Sophia.

Closed limit:

$\sigma + \lambda = \text{const} \iff \zeta_0 = 0.$

▶ [merged] **Genealogy of the gate (Appendix IV).** The earned mode $H_K/(H_s+H_K)$ of g_{eff} and the Mensura kernel of §14 descend from a closed hysteretic prototype. There H_K *amplified* effort; in the open gate it is reassigned to *open reception* to external grace ζ_0. The prototype driver $\eta_H|\dot{u}|\,\text{sp}_m(\cdot)$ was refined into the morally-neutral kernel $\eta\,R((\dot{x}_0)_+)$ ("the measure with which ye mete," Mt 7:2).

[excluded — superseded: the closed $u_{\text{eff}} = u(1 + \rho_H H_K/(H_c + H_K))$ amplifier route and $S_H^{\dagger}, P_H$ — the appendix itself marks these non-canonical; the mature chain is $H_K \to g_{\text{eff}} \to I_{\text{gate}} \to \lambda \to P$.]

13 Sophianic Folding Feedback in the Core

Status of the refinement.

The original Open-Gate core remains the baseline system. Sophianic Folding is added as a conservative refinement: it does not remove the Open-Gate balance, but separates Sophia into an available component and an embodied morphological component. The guiding chain becomes

$$I_{\text{gate}} \to \lambda \to q_{\text{fold}} \to \lambda_{\text{form}} \to P.$$

Sophia does not only accumulate. Once the old Vessel-form becomes insufficient, part of Sophia can be embodied as form.

▷ Plain. Folding is an add-on, not a rewrite: the original equations still hold exactly. It only adds a way for surplus wisdom to set into durable form.

Available Sophia and embodied Sophia.

λ is read as available Sophia, λ_{form} the portion embodied into form: $\lambda_{\text{tot}} = \lambda + \lambda_{\text{form}}$. The original model is recovered by $q_{\text{fold}} = 0,\ Q_{\text{fold}} = 0,\ \lambda_{\text{form}} = 0$.

▷ Plain. Two kinds of wisdom: λ you currently hold, and λ_{form} that has hardened into who you are. Switch folding off and the original model returns untouched.

Pre-fold Sophia production.

Before morphological feedback is applied, the Open-Gate production of Sophia reads: $\dot{\lambda}^{(0)} = -\dot{\sigma} + I_{\text{gate}} = (\delta u - \gamma)\tanh(\kappa\sigma) + I_{\text{gate}}$ — exactly the original Open-Gate law.

▷ Plain. Before any folding, wisdom is produced by exactly the original Open-Gate rule — cleansed resistance plus admitted grace. Nothing new yet; this is the baseline.

Folding activation index and normal form.

Folding switches on once the Sophia-driven index A_{fold} turns positive; q_{fold} then obeys a supercritical pitchfork:

$$A_{\text{fold}} = c_0\lambda + c_1\dot{\lambda}^{(0)} - a_0, \quad B_{\text{fold}} = b + 2c_1\eta_{\text{fold}}\lambda > 0, \quad \dot{q}_{\text{fold}} = A_{\text{fold}}q_{\text{fold}} - B_{\text{fold}}q_{\text{fold}}^3, \quad Q_{\text{fold}} = q_{\text{fold}}^2.$$

On the stable folded branch $Q^*_{\text{fold}} = A_{\text{fold}}/B_{\text{fold}}$ when $A_{\text{fold}} > 0$; if $A_{\text{fold}} < 0$, the old form $q_{\text{fold}} = 0$ remains stable.

▷ Plain. A switch with a spring. While the index A_{fold} is negative the old form is stable; the moment it turns positive a new form appears on its own (a pitchfork) — the mathematics of becoming someone new.

Folding-regulated Sophia flow.

Activated folding re-partitions available Sophia into embodied form without loss:

$$\dot{\lambda} = \dot{\lambda}^{(0)} - \eta_{\text{fold}}Q_{\text{fold}}\lambda, \qquad \dot{\lambda}_{\text{form}} = \eta_{\text{fold}}Q_{\text{fold}}\lambda,$$

so the total balance is preserved: $\frac{d}{dt}(\sigma + \lambda + \lambda_{\text{form}}) = I_{\text{gate}}$, i.e. $\sigma + \lambda + \lambda_{\text{form}} = C_0 + G_{\text{recepta}}$. The apparent loss in $\dot{\lambda}$ is not destruction of Sophia but embodiment of Sophia into the Vessel-form.

▷ Plain. When folding is active, some available wisdom is moved into permanent form. It looks like a loss in λ, but the grand total is conserved — embodiment, not disappearance.

Effective Sophia damping.

On a quasi-stationary folded branch, $\gamma_{\text{eff}} = \gamma_\lambda + \eta_{\text{fold}} Q_{\text{fold}}$; thus $Q_{\text{fold}} > 0 \Rightarrow \dot{\lambda} < \dot{\lambda}^{(0)}$. Folding is the morphological branch of Open-Gate stabilization: it turns excess available Sophia into stable form.

▷ Math. $\dot{q}_{\text{fold}} = A_{\text{fold}} q_{\text{fold}} - B_{\text{fold}} q_{\text{fold}}^3$ is the normal form of a *supercritical pitchfork*: for $A_{\text{fold}} < 0$ only $q_{\text{fold}} = 0$ is stable; as A_{fold} crosses 0 two symmetric branches $q_{\text{fold}} = \pm\sqrt{A_{\text{fold}}/B_{\text{fold}}}$ appear. The induced $-\eta_{\text{fold}} Q_{\text{fold}} \lambda$ is not loss but a conservative re-partition of the Sophia budget into λ_{form}, as the invariant $\sigma + \lambda + \lambda_{\text{form}} = C_0 + G_{\text{recepta}}$ shows.

Core meaning.

$V, F \to u \to \sigma \to \lambda \to q_{\text{fold}} \to \lambda_{\text{form}} \to P$: will becomes action; resistance becomes Sophia; Sophia becomes form; form releases being.

▷ Plain. The whole chain in one breath: will becomes action, action becomes synergy, resistance becomes wisdom, wisdom becomes form, and form releases being.

14 Memory Layer for the Gate

Path coordinate

The path coordinate and its velocity drive the memory dynamics through their positive and negative parts:

$$x_0 = u - \sigma, \quad \dot{x}_0 = \frac{V'F + VF'}{2\sqrt{VF + \varepsilon}} - (\gamma - \delta u)\tanh(\kappa\sigma),$$

$$(\dot{x}_0)_+ = \max(\dot{x}_0, 0), \qquad (\dot{x}_0)_- = \max(-\dot{x}_0, 0).$$

Mensura function

The Mensura saturates its input through a concave Michaelis–Menten law, rising monotonically toward 1:

$$R(z) = \frac{z}{z+K}\,(K > 0, z \geq 0), \quad R(0) = 0,\; R'(z) = \frac{K}{(z+K)^2} > 0,\; R''(z) = -\frac{2K}{(z+K)^3} < 0,\; R \to 1.$$

▷ Plain. A saturating meter: small inputs register almost fully, large ones hit a ceiling near 1. It keeps memory from ever blowing up.

Gate memory and wound memory

Forward motion builds openness H_K; backward motion builds the wound W, each decaying under the other:

$$\dot{H}_K = \eta R((\dot{x}_0)_+) - \nu R((\dot{x}_0)_-)\frac{H_K}{H_s + H_K}, \qquad \dot{W} = \eta\Big[R((\dot{x}_0)_-) - R((\dot{x}_0)_+)\frac{W}{W_s + W}\Big].$$

H_K records acquired openness; W records vulnerability. In the Open-Gate core, H_K opens g_{eff} rather than directly adding to Pleroma. The same-rate prototype is recovered by $\nu = \eta$.

▷ Plain. Moving forward (growth) builds openness H_K; slipping backward builds the wound W. Each gently fades while the other dominates — memory that leans toward healing.

▷ Math. $R(z) = z/(z+K)$ is a saturating (Michaelis–Menten) kernel: concave, $R(0) = 0$, $R \to 1$ — a diminishing-returns measure compressing large movements. $\dot{H}_K$ is a leaky adaptive integrator (gain on forward motion, gated decay on regress); since the gain is ≥ 0, $H_K = 0$ is again an invariant lower boundary.

▶ [merged] **Non-negativity of H_K (Appendix IV).** At $H_K = 0$ the loss term vanishes, so $\dot{H}_K \geq 0$: zero is an absorbing floor — a *Filtrum Tetelestai* for the memory of struggle. (Genealogy note: with the prototype softplus driver, the signed identity $\text{sp}_m(z) - \text{sp}_m(-z) = z$ splits $\dot{H}_K$ into a rich-memory signed integral and a poor-memory one-sided accumulator.)

15 Endogenous Full System

Local definitions

$u = \sqrt{VF + \varepsilon},\ \Lambda_c = \gamma/\delta,\ \mu = 1 - u/\Omega_P.$

Dolorosum

$\rho(u, \sigma) = \sigma(u/\Lambda_c)e^{1-u/\Lambda_c} \geq 0,\ D_V = \rho V,\ D_F = \rho F.$ It attenuates Will and Action multiplicatively, maximal at the threshold $u = \Lambda_c$.

▷ Plain. A suffering term that weighs on both Will and Action, heaviest right at the threshold $u = \Lambda_c$ — the most painful point of the crossing.

▷ Math. $\mu = 1 - u/\Omega_P$ is a logistic (Verhulst) crowding factor: nourishment is throttled as u approaches the carrying capacity Ω_P and reverses past it. Dolorosum $\rho = \sigma(u/\Lambda_c)e^{1-u/\Lambda_c}$ is unimodal in u/Λ_c, peaking exactly at $u = \Lambda_c$ (the maximiser of xe^{1-x}).

Core ODE system

The full Open-Gate dynamics couple Will, Action, resistance, Sophia, and capacity:

$$\begin{cases} \dot{V} = \big(1 - \frac{\sqrt{VF+\varepsilon}}{\Omega_P}\big)(aF + c\lambda) - \rho V, \\ \dot{F} = \big(1 - \frac{\sqrt{VF+\varepsilon}}{\Omega_P}\big)(aV + c\lambda) - \rho F, \\ \dot{\Omega}_P = \alpha\lambda, \quad \alpha > 0, \\ \dot{\sigma} = (\gamma - \delta\sqrt{VF + \varepsilon})\,\tanh(\kappa\sigma), \\ \dot{\lambda} = (\delta\sqrt{VF + \varepsilon} - \gamma)\,\tanh(\kappa\sigma) + I_{\text{gate}}(t), \\ \sigma(t) + \lambda(t) = C_0 + G_{\text{recepta}}(t). \end{cases}$$

▷ Plain. The complete machine: five coupled equations evolving Will, Action, resistance, wisdom, and capacity together. Everything above feeds into these.

Folding-refined core ODE system

(preserves the original equations, adds the folding feedback layer):

$$\begin{cases} \dot{V} = \big(1 - \frac{\sqrt{VF+\varepsilon}}{\Omega_P}\big)(aF + c\lambda) - \rho V, \quad \dot{F} = \big(1 - \frac{\sqrt{VF+\varepsilon}}{\Omega_P}\big)(aV + c\lambda) - \rho F, \\ \dot{\Omega}_P = \alpha\lambda, \quad \dot{\sigma} = (\gamma - \delta\sqrt{VF + \varepsilon})\,\tanh(\kappa\sigma), \\ \dot{\lambda}^{(0)} = (\delta\sqrt{VF + \varepsilon} - \gamma)\,\tanh(\kappa\sigma) + I_{\text{gate}}, \\ A_{\text{fold}} = c_0\lambda + c_1\dot{\lambda}^{(0)} - a_0, \quad B_{\text{fold}} = b + 2c_1\eta_{\text{fold}}\lambda, \\ \dot{q}_{\text{fold}} = A_{\text{fold}}q_{\text{fold}} - B_{\text{fold}}q_{\text{fold}}^3, \quad Q_{\text{fold}} = q_{\text{fold}}^2, \\ \dot{\lambda} = \dot{\lambda}^{(0)} - \eta_{\text{fold}}Q_{\text{fold}}\lambda, \quad \dot{\lambda}_{\text{form}} = \eta_{\text{fold}}Q_{\text{fold}}\lambda, \\ \sigma + \lambda + \lambda_{\text{form}} = C_0 + G_{\text{recepta}}(t). \end{cases}$$

Setting $q_{\text{fold}} = 0$ recovers the original Open-Gate system.

Sophia floor and source-controlled positivity.

$\lambda(t) > -\frac{a}{c}\min\{V, F\}$; equivalently, with $C = \sigma_0 + \lambda_0$, $\sigma(t) < C + \frac{a}{c}\min\{V, F\}$, and in Open-Gate $\sigma(t) < C_0 + G_{\text{recepta}} + \frac{a}{c}\min\{V, F\}$. Inside the live domain this floor prevents the source terms $aF + c\lambda,\ aV + c\lambda$ from starving Will and Action.

▷ Math. The V- and F-equations are *symmetric under the exchange* $V \leftrightarrow F$, so the diagonal $V = F$ is a flow-invariant set. The Sophia-lift $c\lambda$ enters both equally and cancels in the imbalance dynamics, leaving only the asymmetric Dolorosum part — the mechanism of the next section.

Canonical formula chain for Open-Gate grace

Grace flows along a single chain from the gate to Being:

$$I_{\text{gate}} \to \lambda \to \left\{\dot{\Omega}_P = \alpha\lambda,\ \dot{V} = \mu(aF + c\lambda) - \rho V,\ \dot{F} = \mu(aV + c\lambda) - \rho F\right\} \to u \to \{S^\dagger, \Phi\} \to P.$$

With Sophianic Folding active, the same chain acquires a morphological branch:

$$\begin{aligned} I_{\text{gate}} \to \lambda \to \{\Omega_P;\ V, F;\ q_{\text{fold}}\} \to \{&\text{Brim expansion; Will–Action} \\ &\text{nourishment; embodied form}\} \to P. \end{aligned}$$

16 Will–Action Balance

Imbalance

$\Delta(t) = V(t) - F(t)$. **Exact imbalance equation**

$$\frac{d\Delta}{dt} = -(\mu a + \rho)\Delta, \quad \mu = 1 - \frac{\sqrt{VF+\varepsilon}}{\Omega_P}, \quad \rho = \sigma(u/\Lambda_c)e^{1-u/\Lambda_c}.$$

▷ Plain. The gap between Will and Action shrinks over time whenever damping is nonnegative — the system naturally pulls itself toward balance $V = F$.

Exact solution

$\Delta(t) = \Delta_0 \exp\big(-\int_0^t (a\mu + \rho)\, ds\big)$. The manifold $V = F$ is invariant; alignment requires nonnegative damping and a divergent damping integral.

▷ Math. $\dot{\Delta} = -(a\mu+\rho)\Delta$ is *linear* in Δ, giving the exact contraction $\Delta(t) = \Delta_0 \exp(-\int(a\mu+\rho))$ toward the axis $V = F$. Convergence holds iff the damping integral diverges; if the rate decays like t^{-1} the approach is only polynomial, $\Delta \sim C\Delta_0/t$ — the marginal case of the Law of Ontological Integrity.

▶ [merged] **Law of Ontological Integrity.** In the symmetric source, the Sophia-lift $c\lambda$ *cancels* (it feeds V and F equally), while Dolorosum does not, leaving the compressive $-\rho\Delta$. When μ, ρ decay only as t^{-1}, convergence is conditional and polynomial: $\Delta(t) \sim C\Delta_0/t$. Integrity is held not by force but by the symmetry of nourishment.

17 Brim Expansion and Epektasis

Brim law

$\dot{\Omega}_P = \alpha\lambda,\ \Omega_P(t) = \Omega_P(0) + \alpha \int_0^t \lambda$. Sophia expands the capacity of the Vessel; Epektasis is sustained expansion rather than a final static maximum.

▷ Plain. Capacity grows in step with accumulated wisdom: the more Sophia you gather, the larger the vessel you can hold. This endless widening is Epektasis.

Asymptotic Sophia.

If the received-grace integral converges and $\sigma \to 0$, then $\lambda \to \lambda_\infty = \sigma_0 + \lambda_0 + G_{\text{recepta}}(\infty)$. If $G_{\text{recepta}}(\infty) = +\infty$, unbounded Sophia is possible in the open model.

Epektasis with folding stabilization.

Folding does not stop Epektasis; the Brim law remains $\dot{\Omega}_P = \alpha\lambda$; what changes is the available flow $\dot{\lambda} = \dot{\lambda}^{(0)} - \eta_{\text{fold}} Q_{\text{fold}} \lambda$. *Open-Gate expands; Folding forms.*

Pleroma Christi condition.

$\int_0^\infty \lambda = +\infty \Rightarrow \Omega_P \to +\infty$; if additionally $\lambda \to \lambda_\infty > 0$, then $\Omega_P \sim \alpha\lambda_\infty t$. The internal cleansing signature is $\sigma \to 0,\ S^\dagger \to E_0$: the Brim expands through sustained Sophia, and the visible cleansing signature is the disappearance of singular resistance and the saturation of Gratia Synergica.

▷ Plain. If wisdom keeps flowing forever, capacity grows without bound and resistance vanishes — the visible signature of full cleansing: $\sigma \to 0$, grace saturated.

▷ Math. Ω_P is a *pure integrator* of Sophia, $\Omega_P = \Omega_P(0) + \alpha \int \lambda$, with no fixed point unless $\lambda \to 0$. A persistent $\lambda \to \lambda_\infty > 0$ forces unbounded linear growth $\Omega_P \sim \alpha\lambda_\infty t$: there is *no static maximum* and no attractor in Ω_P — that absence *is* Epektasis.

▶ [merged] **Pleroma Christi — Theosis, non-attractor, conditional stability.** Pleroma Christi is the absorbing state w.r.t. σ but *not* the final attractor w.r.t. Ω_P: motion continues within it without end. It is the asymptotic regime of inexhaustible Fullness, not a capacity. Theosis is deification — full correspondence of Image and Likeness, *not* a fusion in which the person disappears. Stability is conditional: $\delta\sqrt{VF + \varepsilon} \geq \gamma$ stable, $< \gamma$ unstable — not repose, but flawless continuous maintenance of fullness (John 1:14; Eccl 1:7, the sea is never full).

18 Live Domain and Death Boundary

Live domain

Life is the open region where both faculties are positive and synergy stays within capacity:

$$\mathcal{D}_{\text{live}} = \{(V, F, \sigma, \lambda, \Omega_P) : V > 0, F > 0, \sigma \geq 0, \Omega_P > \Lambda_c, 0 < u < \Omega_P\}.$$

The live domain is the region where the continuous V.F.S. dynamics are admissible.

▷ Plain. The region where you are truly alive: both Will and Action positive, capacity above threshold, synergy still inside capacity.

Death boundary

Death is reached when a faculty vanishes, capacity collapses to threshold, or synergy saturates capacity:

$$\partial\mathcal{D}_{\text{death}} = \{V = 0\} \cup \{F = 0\} \cup \{\Omega_P \leq \Lambda_c\} \cup \{u \geq \Omega_P\}, \qquad X_{\text{VFS}} : \mathcal{D}_{\text{live}} \to T\mathcal{D}_{\text{live}},$$

$$t_{\text{death}} = \inf\{t > 0 : x(t) \notin \mathcal{D}_{\text{live}}\}.$$

The canonical continuous flow is defined only on the living interior. Resurrection, if present, is not a smooth term inside the vector field but a hybrid re-entry map after boundary contact.

▷ Plain. The walls of life: a faculty hitting zero, capacity collapsing to the threshold, or synergy outgrowing its vessel. Touching any wall is death.

▷ Math. $\mathcal{D}_{\text{live}}$ is an open set and t_{death} a first-exit time, so the model is a *hybrid (impulsive) dynamical system* — continuous flow on $\mathcal{D}_{\text{live}}$, punctuated by a discrete reset on $\partial\mathcal{D}_{\text{death}}$ — not a single smooth ODE.

19 ▶ Lyapunov Stability of the Active Domain (summary)

[merged]

A single dissipative Lyapunov functional certifies the active domain (weights renamed to avoid the 2.0 $A = d^2P/dt^2$):

$$\mathcal{L}_{pb} = w_\sigma \Psi_\sigma + w_\Delta D_\Delta + w_q q^2.$$

Results: $\mathcal{D}_A$ positively invariant; $\mathcal{L}_{pb}$ dissipative; the death-boundary is never reached ($h_1, h_2 \geq m_{pb} > 0$); resets are uniformly non-Zeno ($\Delta t_j \geq \Delta t_* > 0$); $\sigma \to 0$ and imbalance $\to 0$ exponentially; the product-balance mechanism needs no artificial $q \geq q_0$. Two bridges: hybrid resets *are* the Resurrection operator $\mathfrak{R}$, and a resurrection must *cleanse, not worsen* ($\Psi_\sigma(x^+) \leq \Psi_\sigma(x^-),\ D_\Delta(x^+) \leq D_\Delta(x^-)$). This is not a global claim (collapse persists outside $\mathcal{D}_A$); $\sigma \to 0$ is privatio boni. *(Full proof: Lyapunov volume; here in summary only.)*

20 Resurrectio: Hybrid Re-entry Map

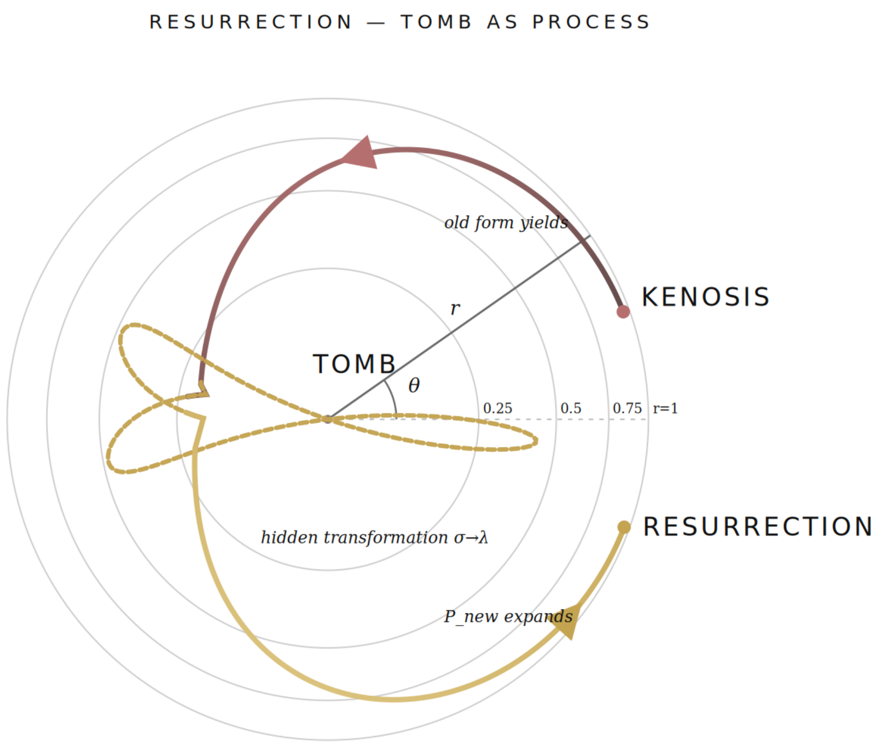

Figure 7: Resurrection as process: kenosis (old form yields) descends to the tomb — the hidden transformation $\sigma \to \lambda$ — whence resurrection expands (P_{new} grows).

This is included because it appears in `index.html`; it is not taken from the appendix or Lyapunov file. **Reset symbols:** Γ_R resurrection-mode grace, H_R resurrection memory, D_{death} depth of death-boundary contact; $\Theta_\bullet, \beta_\bullet, \chi_\bullet, \eta_R, \theta_\Gamma, \alpha_\Omega, \beta_\Omega, \omega_R \geq 0$ are reset coefficients, not continuous live-flow coefficients.

▷ Plain. The big picture. Everywhere else life is a smooth flow. Here is the one exception: death means the state has left the living region, and resurrection is not gradual healing but a single instantaneous jump back in. That is why this section uses a one-step reset rule $\mathfrak{R}$ instead of a differential equation.

Resurrection admissibility

Re-entry is admissible only when the weighted resurrection score clears the critical bar:

$$\mathcal{A}_R = \Theta_\Gamma \Gamma_R + \Theta_\Phi \Phi_R + \Theta_\lambda \lambda^+ + \Theta_H H_R - \Theta_\sigma \sigma - \Theta_D D_{\text{death}}, \quad \mathcal{A}_R > R_c \iff \text{re-entry admissible.}$$

▷ Plain. Picture a doorman with a scorecard. Grace Γ_R, inner light Φ_R, available Sophia λ^+ and memory H_R all add points; leftover resistance σ and the depth of death D_{death} subtract them. Beat the pass-mark R_c and you are let back in.

Participated resurrection grace

$\Gamma_R^{\text{human}} = \pi \Gamma_{\text{Christi}}, \; \pi \in [0, 1].$

▷ Plain. The grace that raises you is not self-produced. It is a share of Christ's resurrection-grace: $\pi = 0$ means none received, $\pi = 1$ means received in full.

Hybrid re-entry map

$\mathfrak{R} : \partial\mathcal{D}_{\text{death}} \to \mathcal{D}_{\text{live}}$, $x(t^+_{\text{death}}) = \mathfrak{R}(x(t^-_{\text{death}}))$:

$$
\begin{aligned}
&V^+ = V_R + \beta_V \Gamma_R + \chi_V H_R, \quad F^+ = F_R + \beta_F \Gamma_R + \chi_F H_R,\\
&\sigma^+ = q_R \sigma^- \ (0 \le q_R < 1), \quad \lambda^+ = \lambda^- + (1-q_R)\sigma^- + \eta_R H_R + \theta_\Gamma \Gamma_R,\\
&\Omega_P^+ = \Omega_R + \alpha_\Omega \Gamma_R + \beta_\Omega \lambda^+ + \chi_\Omega H_R, \quad H_R^+ = H_R^- + \omega_R D_{\text{death}},
\end{aligned}
$$

with $V_R, F_R > 0$, $\Omega_R > \Lambda_c$, all reset coefficients ≥ 0. **Landing condition** $\Omega_R + \alpha_\Omega \Gamma_R + \beta_\Omega \lambda^+ + \chi_\Omega H_R > \sqrt{V^+F^+} + \varepsilon$: the reset must land inside a live capacity large enough to contain the new synergic intensity.

▷ Plain. This is the jump itself. Will and Action are re-seeded to fresh positive values; resistance shrinks ($\sigma^+ = q_R\sigma^-$); the removed part is poured into Sophia; capacity is reset wide enough to hold the new life; and memory records how deep the death was. Genuinely new, yet remembering.

▷ Math. $\mathfrak{R}$ is the discrete transition of the hybrid automaton: $\sigma^+ = q_R\sigma^-$ contracts resistance ($0 \le q_R < 1$) while λ^+ absorbs the cleansed part plus grace and memory. Admissibility $\mathcal{A}_R > R_c$ is the *guard*, and the landing condition keeps the image in $\mathcal{D}_{\text{live}}$ so the execution continues; each arc carries its own first integral ($C_{j+1} \neq C_j$).

Open balance across life arcs

$\frac{d}{dt}(\sigma + \lambda) = \zeta_0 g_{\text{eff}} e^{-\phi\sigma}(1 + \chi\lambda_+)^{-1} \ge 0$ inside each arc; $\sigma + \lambda$ jumps under $\mathfrak{R}$ ($C_{j+1} \neq C_j$).

▷ Plain. Within one lifetime the running total resistance-plus-Sophia only grows. At resurrection the books are re-opened and the total jumps — continuity inside a life, a clean break between lives.

▶ [merged] **Una Gratia, Duo Modi.** Mode I = continuous gate inflow in life; Mode II $= \Gamma_R$, gate-free transformation at the death boundary — one grace theologically, two operators mathematically. *Recovery* (continuous healing inside $\mathcal{D}_{\text{live}}$) must never be conflated with *resurrection* (discontinuous re-entry after leaving it). H_R makes it "a new beginning with memory, not a reset."

▶ [merged] **Depth of death** (softplus sharpness β_D, distinct from the reset $\beta_\bullet$): $D_{\text{death}} = \text{sp}_{\beta_D}(-\lambda(t_{\text{death}})) \ge 0$, $\text{sp}_{\beta_D}(z) = \frac{1}{\beta_D}\ln(1 + e^{\beta_D z})$. At death $\Phi_R \approx 0$, so admissibility rests on grace Γ_R and memory H_R — *grace, not merit*. Memory may resemble a karmic trace, but the engine is grace; the telos is Fullness, not a wheel of rebirth, and the wound W is to be healed, not an account to be paid.

21 ▶ Theological–Mathematical Synopsis [merged]

Object	**Theological + analytical content**
$u = \sqrt{VF + \varepsilon}$	Manere — abiding synergy of will and action
$\Phi(x)$	Filtrum Lucis — light in the darkness ($\Phi > 0$)
E_0	Imago Dei / Mustard Seed — the non-dynamic ground
$g_{\text{eff}}, I_{\text{gate}}$	Open-Gate grace — received, not generated
$S^\dagger$	Gratia Synergica — grace responding to effort
$P = E_0 + S^\dagger + \Phi$	Dynamic Fullness — Imago + grace + passage
$\dot{V}, \dot{F} \Rightarrow V{=}F$	integrity — will and action aligned
$\sigma + \lambda = C_0 + G$	the balance of resistance and Sophia under grace
$\delta u \geq \gamma$	surgery threshold — transformation over collapse
S, A, K, An	Gignesthai $\to$ Odinai $\to$ Katharsis $\to$ Anastasis
$\Omega_P \to \infty$	Epektasis — inexhaustible expansion

Canonical reading: *V.F.S. is an Open-Gate dynamical theology.*

22 Compact Core Summary

$$V + F \to u \to x = u - \sigma \to \Phi(x) \to P(t).$$

$$\sigma \to \text{Tetelestai} \to \text{Katharsis} \to \text{Anastasis} \to \lambda.$$

$$\text{Open Gate} \to I_{\text{gate}} \to \lambda \to \Omega_P \to \text{Epektasis}.$$

$$\lambda \to q_{\text{fold}} \to \lambda_{\text{form}} \;\equiv\; \text{Sophia becomes Vessel-form}.$$

$$\dot{\lambda} = \dot{\lambda}^{(0)} - \eta_{\text{fold}} Q_{\text{fold}} \lambda, \quad \dot{\lambda}_{\text{form}} = \eta_{\text{fold}} Q_{\text{fold}} \lambda, \quad \sigma + \lambda + \lambda_{\text{form}} = C_0 + G_{\text{recepta}}(t).$$

$$P \to S \to A \to K \to An \;\equiv\; \text{Gignesthai} \to \text{Odinai} \to \text{Katharsis} \to \text{Anastasis}.$$

Revision Note

This expanded version adds the Paschal Triad of Filtrum Lucis, the symmetry of zeros around the Cross, the fourth-derivative support formulas needed for $An(t)$, the Sophia floor, explicit death-boundary definitions, bounded gate inflow, the Human Balance identity for $\dot{u}$, asymptotic Sophia, reset-symbol clarification, and the distinct memory-forgetting rate ν. This hybrid version preserves the existing structure and original core formulas, while adding Sophianic Folding as an explicit Open-Gate refinement: available Sophia may become embodied Sophia through a Lyapunov-compatible morphological feedback branch. The original index-only core is recovered by setting $q_{\text{fold}} = 0$.

▸ [merged] **Patch merge.** Added on top of the unchanged core: the Laws of Unified Being (I,II,III,V) with Trinitarian analogies; the Theological–Mathematical Synopsis; the two-level synergy and the "not holiness" caveat; the Genealogy of the gate and the H_K non-negativity lemma; the curvature portrait and honesty caveats of the Paschal triad; the Law of Ontological Integrity; Pleroma Christi as Theosis / non-attractor / conditional stability; the Lyapunov summary; and Resurrectio's *Una Gratia, Duo Modi* with D_{death}. Deliberately *excluded* as superseded: the Fading-Impulse $\mathcal{P}(t)$/First-Call term and the closed u_{eff} amplifier route. All merged symbols follow the 2.0 convention; nothing original was removed.